

Prognosis and Clinical Course

If left untreated, the condition can persist for many years. In immunocompetent individuals, the number of mites may decrease over time.

▲ FIGURE 208-4 Crusted scabies. Hyperkeratotic plaques populated with thousands of mites.

▲ FIGURE 208-5 Scabies. Microscopic examination of a mineral oil preparation after scraping a burrow reveals a gravid female mite with oval, gray eggs and fecal pellets.

Treatment

There are a number of treatments for scabies with varying levels of effectiveness. Factors influencing treatment choice include patient age, medication cost, severity of eruption, and prior treatment success.

In adults, topical scabicides should be applied to the entire skin surface except the face and scalp, with careful attention to intertriginous areas, genitalia, periungual regions, and behind the ears. In children and patients with crusted scabies, the face and scalp should also be treated.

Patients should be informed that the rash and pruritus may persist for up to 4 weeks after appropriate scabicial therapy. This post-scabies pruritus does not indicate treatment failure and may lead to inappropriate overuse of medications if not properly explained. Topical steroids, antihistamines, and, if needed, a short course of systemic steroids can be used to manage persistent pruritus and inflammation after completion of scabicide treatment. The treatment of scabies is summarized in Box 208-2.

Box 208-1

Differential Diagnosis of Scabies

Most Likely

- Atopic dermatitis
- Insect bite reactions
- Contact dermatitis
- Dermatitis herpetiformis
- Dyshidrotic eczema

Consider

- Psoriasis, particularly in the crusted variety
- Bullous pemphigoid, when vesicles and bullae are present
- Drug eruption

Ivermectin

The only oral but highly effective scabicide currently available is **ivermectin**. It was discovered in the 1970s from the fermentation broth of the soil actinomycete *Streptomyces avermitilis*. Structurally similar to macrolide antibiotics, it lacks antibacterial activity.

Ivermectin is currently approved by the U.S. Food and Drug Administration (FDA) for the treatment of intestinal stage onchocerciasis and strongyloidiasis, but it is not FDA-approved for scabies. Its action against scabies mites is due to its high affinity for glutamate-gated chloride channels in the peripheral nervous systems of invertebrates. By enhancing chloride ion influx, ivermectin

disrupts nerve and muscle cell function, leading to paralysis and death of the parasite.

In mammals, these receptors are primarily located in the central nervous system, and ivermectin normally does not cross the blood-brain barrier. However, conditions that compromise the blood-brain barrier may allow CNS entry, increasing the risk of neurotoxicity.

Therefore, ivermectin should generally be avoided in children under 5 years of age or weighing less than 15 kg, and during pregnancy or lactation. Nonetheless, there are reports of its use in these populations without adverse effects.

Numerous studies have demonstrated the effectiveness of ivermectin in treating scabies. The typical dosage is **200 µg/kg**, often repeated after 10 to 14 days, although the optimal regimen has not been definitively established.

A clinical trial comparing ivermectin to topical 5% permethrin showed that a single dose of ivermectin achieved a 70% cure rate, increasing to 95% with a second dose. In contrast, a single application of permethrin resulted in a cure rate of nearly 98%.

A topical formulation of 1% ivermectin has shown promise in early studies but is not yet commercially available.